

# CIFRIS26 · Programma

Aula Magna

Torre A, 2° piano · Università Roma Tre

Lunedì 5 ottobre 2026

## 08:30–09:00 · Registrazione e consegna badge

La sessione di apertura è introdotta da Germana Zuffanti  
(*Divisione Comunicazione e Rapporti Istituzionali, ACN*).

## 09:00–09:15 · Saluti

Prof. Massimiliano Fiorucci (*Rettore, U. Roma Tre*)

Amm. Luigi Schinelli (*Capo Servizio Crittografia, ACN*)

Prof. Massimiliano Sala (*Presidente De Cifris*)

## 09:15–09:30 · Le attività di De Cifris

Marco Pedicini (*Direttore Scientifico De Cifris*)

## 09:30–09:45 · Keynote

Santo Mirabelli (*Polizia Postale*)

Crittografia: la sfida dell'accesso legittimo ai dati nell'era delle comunicazioni cifrate

## 09:45–10:15 · De Cifris Partners

09:45–09:50 Andrea Gelpi (*coordinatore di De Cifris Partners*)

09:50–09:55 Francesco Barbaro (*BANCOMAT S.p.A.*)

09:55–10:00 Federico Lombardi (*Conio S.r.l.*)

10:00–10:05 Mirko Frighetto (*DataKrypto S.r.l.*)

10:05–10:10 Massimo Caccia (*Random Power S.r.l.*)

10:10–10:15 Tiziana Liberati (*E4 Computer Engineering S.p.A.*)

## 10:15–11:00 · Panel ACN

Modera: Roberto Civino (*Divisione Algoritmi e Applicazioni Crittografiche, ACN*)

Costruire la filiera crittografica nazionale, dalla ricerca all'industria

Marco Baldi (*UNIVPM*)

Francesca Galli (*Segreteria Tecnica, MUR*)

Marco Mangiulli (*Capo Ricerca e Sviluppo, ARUBA S.p.A.*)

## 11:00–11:30 · Coffee break

The opening session will be conducted in Italian.

# Scientific Sessions

Aula Magna (Tower A, 2nd Floor)

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**Session I chair:** Roberto La Scala (*U. Bari*)

**11:30–12:00**     **Marco Macchetti** (*Qualcomm Wireless GmbH*)  
The challenges of securing PQC on embedded systems

**12:00–12:30**     **Christopher Battarbee** (*U. Edinburgh*)  
The rise and fall of semidirect product cryptography in finite groups

**12:30–13:00**     **Elena Andreeva** (*TU Wien*)  
Expanding primitives, expanding possibilities in symmetric cryptography

**13:00–14:00**     **Lunch break**

**Session II chair:** Alessandro Neri (*U. Federico II, Napoli*)

**14:00–14:30**     **Violetta Weger** (*TUM*)  
Where does code-based cryptography stand, and what will the future bring?

**14:30–15:00**     **Léo Perrin** (*Inria*)  
The Cambrian Explosion of Symmetric Techniques for Advanced Protocols

**15:00–15:30**     **Paola de Perthuis** (*CWI*)  
Recent advances in module-lattice reduction models

**15:30–15:45** · **Coffee break**

**Following · Parallel workshops**

**15:45–19:00**     Parallel workshops (TAC from 16:15; NT from 17:30)

**17:15–17:30** · **Coffee break**

## Recent Advances in Post-Quantum Cryptography

- 15:45–16:30**   **Iliaria Zappatore** (*U. Limoges*)  
Decoding Gabidulin codes and their generalisations  
with application to constant-time RQC
- 16:30–17:15**   **Guido Lido** (*U. Roma Tor Vergata*)  
A generalisation of SQISign
- 17:15–17:30** · **Coffee break**
- 17:30–18:15**   **Laura Mattiuz** (*UNIVPM*)  
On the hardness of finding permutation  
automorphisms of linear codes
- 18:15–19:00**   **Edoardo Signorini** (*European Commission JRC*)  
Efficient straight-line extractable NIZKPs for  
cryptographic group actions

## Organising Committee

Marco Baldi (*UNIVPM*), Alessio Caminata (*U. Genova*), Annamaria Iezzi (*Université Grenoble Alpes*), Alessio Meneghetti (*U. Bari*), Edoardo Persichetti (*Florida Atlantic University*), Federico Pintore (*U. Trento*).

## Topics in Applied Cryptography

- 16:15–16:45**   **Andrea Gangemi** (*ShunyaXifra*)  
On delegation of verifiable presentations from mdoc and BBS credentials
- 16:45–17:15**   **Simone Dutto** (*ACN*)  
The cost of quantum resistance: comparing classical and post-quantum cryptography
- 17:15–17:30** · **Coffee break**
- 17:30–18:00**   **Alessandro Barengi** (*PoliMi*)  
Tough cookies in the PQC migration: where current standards will have a hard time chewing
- 18:00–18:30**   **Alessandra Dolmeta** (*PoliTo*)  
Beyond the algorithm: Hardware/Software Design for post-quantum cryptography

**TAC local chair:** Silvia Onofri (*Scuola Normale Superiore*).

TAC Organising Committee on page 7.

<https://aleph.fbk.eu/events/TAC2026>

# OPEN · NT

DAMS, Via Ostiense 139 (raised ground floor)

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## Open Workshop

Aula 3bis

- 15:45–16:05** **Silvia Onofri** (*Scuola Normale Superiore, Pisa*)  
Quantum Private Queries in Entanglement Routing Protocols
- 16:05–16:25** **Gabriele Radici** (*U. Trento*)  
Binary quadratic systems and their relation to complexity theory
- 16:25–16:45** **Leonardo Errati** (*PoliTo*)  
Canonical liftings: beyond signature compression
- 16:45–17:00** **Ester Diago et al.** (*U. Trento*)  
A blockchain-based quiz game
- 17:15–17:30** · **Coffee break**
- 17:30–17:50** **Tiziana Liberati** (*E4 Computer Engineering*)  
GPU acceleration of Learning With Errors KEMs using OpenACC for post-quantum cryptography
- 17:50–18:10** **Aurora Pistocchio et al.** (*U. Trento*)  
Quantum risks in public blockchains: exposure, readiness, and migration strategies
- 18:10–18:30** **Lorenzo Viganò** (*U. Trento*)  
Sparkle+ and its security claims
- 18:30–18:45** **Edoardo Barbieri et al.** (*U. Trento*)  
Scalable and verifiable off-chain gaming tournaments on the XRP Ledger via Zero-knowledge proofs
- Chair:** Massimo Caccia (*Random Power*).

## Number Theory and Cryptography

Aula 5

- 17:30–18:10** **Daan van Gent** (*U. Leiden*)  
Attacks on module lattices using (Galois) symmetries
- 18:10–18:25** **Alessandro Cobbe** (*UniBw München*)  
The Rabin cryptosystem over number fields
- 18:25–18:40** **Nadir Murru** (*U. Trento*)  
Norms and Embeddings in Fully Homomorphic Encryption
- 18:40–18:55** **Francesco Sica** (*FAU*)  
Improvements to multi-scalar multiplication for zero-knowledge arguments
- NT Organising Committee:** Andrea Gangemi (*ShunyaXifra*), Pietro Mercuri (*U. Trento*).

## Symmetric and Code-Based Cryptography

**15:45–16:30** **André Schrottenloher** (*Inria Rennes*)

Recent advances in quantum algorithms for cryptanalysis

**16:30–17:15** **Valentino Smaldore** (*U. Padova*)

Three-weight rank-metric codes and regular fat linear sets

**17:15–17:30** · **Coffee break**

**17:30–18:15** **Emanuele Bellini** (*TII*)

Automated cryptanalysis in symmetric cryptography: CLAASP features and tool comparison

**18:15–19:00** **Roberto Civino** (*ACN*)

Circle-linear cryptanalysis: Bibrace characters and weak-key linear distinguishers for CRAFT

## Chairs

**Alessandro Casale** · *ACN*

**Marco Pedicini** · *Roma Tre University*

**Massimiliano Sala** · *De Cifris*

## Event coordination

cifris26@decifris.it (Valentina)

## TAC Organising Committee

**Riccardo Longo (chair)**

*Fondazione Bruno Kessler, Center for Cybersecurity*

**Veronica Cristiano**

*Telsy*

**Marco Pedicini**

*Roma Tre University*

**Silvio Ranise**

*Fondazione Bruno Kessler, Center for Cybersecurity*

*University of Trento, Department of Mathematics*

**Alessandro Tomasi**

*Fondazione Bruno Kessler, Center for Cybersecurity*

**Stefano Berlato**

*Fondazione Bruno Kessler, Center for Cybersecurity*

**Chiara Spadafora**

*Swiss Post*

## TAC local chair

Silvia Onofri · *Scuola Normale Superiore*

The PQ, SYMM, and NT committees are listed on their respective programme pages.



## Roma Tre

### The venue of CIFRIS26

The conference takes place at the **University Rectorate**,  
Via Ostiense 133B:

**Aula Magna** (Tower **A**, 2nd Floor) — plenary and scientific sessions, PQ workshop

**Hall C** (Tower **C**, hall) — TAC workshop

**Aula 2** (Building **D** — Piazza Telematica — 1st floor) — SYMM workshop

The workshops also take place at the **DAMS building**,  
Via Ostiense 139:

**Aula 3bis** (raised ground floor) — OPEN workshop

**Aula 5** (raised ground floor) — NT workshop

### Coffee break

11:00–11:30 · 15:30–15:45 · 17:15–17:30